

CSC361

Artificial Intelligence

AI Recipe Generator – Report

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1. Introduction

The AI Recipe Generator is a Python-based GUI application that allows users to generate creative and well-structured recipes based on the ingredients they provide. The program uses the Google Gemini API to process user input and produce human-like, coherent recipe text that includes a title, introduction, ingredients, and cooking steps. It features an intuitive Tkinter graphical interface where users can type ingredients, select a meal type, and view the generated recipe inside a scrollable text box. Additionally, it includes built-in error handling that alerts users if no ingredients are entered or if an API connection issue occurs, ensuring smooth and user-friendly operation.

2. Implementation Details

Programming Language: Python

Libraries Used:

- requests: to send HTTP requests and interact with the Gemini API
- tkinter: for creating the graphical user interface
- ttk: for modern UI widgets
- scrolledtext: to display long recipe outputs

How It Operates

1. User Input Handling:

The user types a list of ingredients into the input field.

If the input is empty, a warning message appears to notify the user to enter at least one ingredient.

2. Meal Type Selection:

The user selects a meal type (Main Course, Dessert, Salad, Soup, Snack, or Breakfast) from a dropdown list.

3. API Call:

The application sends a structured JSON request to the Gemini 2.5 Flash API with the selected ingredients and meal type.

The API then returns a structured and coherent recipe in text form.

4. Recipe Display:

The result is displayed in a scrollable text box (ScrolledText) that shows the title, introduction, ingredients list, and cooking steps.

5. Error Handling:

The program catches invalid inputs and API connection issues using try-except blocks, ensuring the app does not crash unexpectedly.

6. User Interface:

The GUI uses warm brown and beige colors for a cooking-inspired aesthetic and includes a single “Generate Recipe” button for simplicity.

3. Challenges Faced and Solutions

Challenge	Solution
Connecting to the Gemini API	Used the official Google Generative Language endpoint and handled authentication with x-goog-api-key.
Managing API errors	Implemented try-except blocks to show clear error messages using messagebox.showerror().
Ensuring readable output	Used the ScrolledText widget to format and display long recipe text neatly.
Designing a user-friendly UI	Applied ttk widgets and warm background colors for a modern and pleasant appearance.

4. Example Runs

Example 1

- Meal Type: Salad
- Ingredients: lettuce, tomato, cucumber, olive oil, lemon

AI Recipe Generator

Enter Ingredients:

lettuce, tomato, cucumber, olive oil, lemon

Select Meal Type:

Salad

Generate Recipe

Generated Recipe

Recipe Title: Sun-Kissed Garden Salad

****Introduction:****
Embrace the essence of fresh, vibrant flavors with this simple yet utterly delicious Sun-Kissed Garden Salad. Featuring crisp lettuce, juicy tomatoes, cool cucumber, and a bright lemon-olive oil dressing, it's a refreshing side dish or light meal perfect for any occasion. It's incredibly easy to prepare and highlights the natural goodness of each ingredient.

Ingredients:

- * 1 head of lettuce (such as romaine, butter, or green leaf), washed and dried
- * 2 ripe medium tomatoes, washed
- * 1 large cucumber (English or regular), washed

AI Recipe Generator

Enter Ingredients:

lettuce, tomato, cucumber, olive oil, lemon

Select Meal Type:

Salad

Generate Recipe

highlights the natural goodness of each ingredient.

Ingredients:

- * 1 head of lettuce (such as romaine, butter, or green leaf), washed and dried
- * 2 ripe medium tomatoes, washed
- * 1 large cucumber (English or regular), washed
- * 3 tablespoons extra virgin olive oil
- * 1 tablespoon fresh lemon juice (from about half a lemon)
- * Salt to taste (recommended: 1/4 teaspoon)
- * Freshly ground black pepper to taste (recommended: 1/8 teaspoon)

Cooking Steps:

1. ****Prepare the Vegetables:**** Tear or chop the lettuce into bite-sized pieces and place them in a large salad bowl. Dice the tomatoes into approximately 1-inch pieces and add them to the bowl. Slice the cucumber into half-moon shapes or dice it, then add to the bowl with the other vegetables.

AI Recipe Generator

Enter Ingredients:

lettuce, tomato, cucumber, olive oil, lemon

Select Meal Type:

Salad

Generate Recipe

Cooking Steps:

1. ****Prepare the Vegetables:**** Tear or chop the lettuce into bite-sized pieces and place them in a large salad bowl. Dice the tomatoes into approximately 1-inch pieces and add them to the bowl. Slice the cucumber into half-moon shapes or dice it, then add to the bowl with the other vegetables.
2. ****Make the Dressing:**** In a small separate bowl, whisk together the extra virgin olive oil, fresh lemon juice, salt, and black pepper until well combined. Taste and adjust seasoning if necessary, adding more lemon for tang or more salt/pepper as desired.
3. ****Dress and Toss:**** Pour the prepared dressing evenly over the vegetables in the salad bowl.
4. ****Serve:**** Gently toss the salad to ensure all the vegetables are lightly coated with the dressing. Serve immediately and enjoy the fresh flavors!

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☒ Recipe generation complete.

Example 2

- Meal Type: Dessert
- Ingredients: milk, sugar, flour, strawberries

AI Recipe Generator

Enter Ingredients:

milk, sugar, flour, strawberries

Select Meal Type:

Dessert

Generate Recipe

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Generated Recipe

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Recipe Title: Strawberry Cloud Delight

Introduction:

Indulge in the simple pleasure of homemade comfort with our Strawberry Cloud Delight. This exquisite dessert brings together the creamy richness of a classic stovetop pudding with the vibrant, fresh sweetness of ripe strawberries. It's a surprisingly easy recipe that promises a delightful, silky texture and bright flavors, making it a perfect treat for any occasion, from a casual family dinner to a comforting individual indulgence.

Ingredients:

AI Recipe Generator

Enter Ingredients:

milk, sugar, flour, strawberries

Select Meal Type:

Dessert

Generate Recipe

Ingredients:

* **For the Pudding:**

* 3 cups (720ml) whole milk

* 1/2 cup (100g) granulated sugar, divided

* 1/4 cup (30g) all-purpose flour

* **For the Topping:**

* 2 cups fresh strawberries

Cooking Steps:

1. **Prepare the Strawberries:** Wash the strawberries thoroughly. Hull them (remove the green tops) and slice them into halves or quarters, depending on their size. Place the sliced strawberries in a small bowl, sprinkle with 1 to 2

AI Recipe Generator

Enter Ingredients:

milk, sugar, flour, strawberries

Select Meal Type:

Dessert

Generate Recipe

5. ****Cool and Chill:**** Remove the pudding from the heat. Pour the warm pudding into individual serving bowls or a larger dish. Allow it to cool slightly at room temperature for about 15-20 minutes. For the best texture and flavor, cover the bowls or dish with plastic wrap (pressing it directly onto the surface of the pudding to prevent a skin from forming) and chill in the refrigerator for at least 1 hour, or until completely cold. The pudding will thicken further as it cools.

6. ****Assemble and Enjoy:**** Just before serving, spoon the macerated strawberries (along with any delicious, sugary juices they've released) generously over the chilled pudding. Serve immediately and savor your homemade Strawberry Cloud Delight!

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☒ Recipe generation complete.